

제 7회 전자구조계산 여름학교

실습 1: EDISON 홈페이지에서의 양자계산 패키지 설치 (VS Code)

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1. Edison homepage
 - a. 계정 만들기
 - b. 다양한 App 활용
 - Crystal Viewer, File Browser, Jupyter Notebook, VS Code
2. VS Code
 - a. 실행
 - b. Panel
 - Editor, Terminal, ...
 - c. Job
 - Submit
 - 확인, ...
3. Linux
 - a. Linux 환경
 - b. Command, variable, etc
 - ls
 - git
 - Bash, ...
4. 수업
5. Atom install

1. Edison homepage

a. Make an account

- Homepage: <https://nanornd.edison.re.kr/home>

The screenshot shows the top navigation bar of the Edison homepage. The navigation items are: Home, About, Data, AI, Service, Tool, IDE, Forum, Curriculum, and Support. The 'Curriculum' item has a dropdown menu open, showing 'ENPC' and 'MSDP'. The 'Service' item has a dropdown menu open, showing a list of services. The 'IDE' item has a dropdown menu open, showing a list of IDEs. The 'Forum' item has a dropdown menu open, showing 'Forum' and 'Q & A'. Red circles are drawn around the 'Service', 'IDE', 'Forum', and 'Curriculum' items. Red arrows point from these circles to the corresponding dropdown menus.

Curriculum ▾

- ENPC
- MSDP

Home About ▾ Data ▾ AI ▾ Service ▾ Tool ▾ IDE ▾ Forum ▾ Curriculum ▾ Support ▾

1st Designer

Data Analyzer

Energy Prediction

Voltage Prediction

Job Monitoring

[ai] DARWIN: Science ChatBot

[ai] DePlot: Plot-to-Table Trans

Graph Generator

Crystal Editor

Crystal Viewer 1.0 (beta)

Crystal Viewer (url link test)

Workbench

Jupyter Notebook

Kernel Manager (simple)

Kernel Manager (advanced)

File Browser (beta)

VS Code (beta)

URL Translator

Forum

Q & A

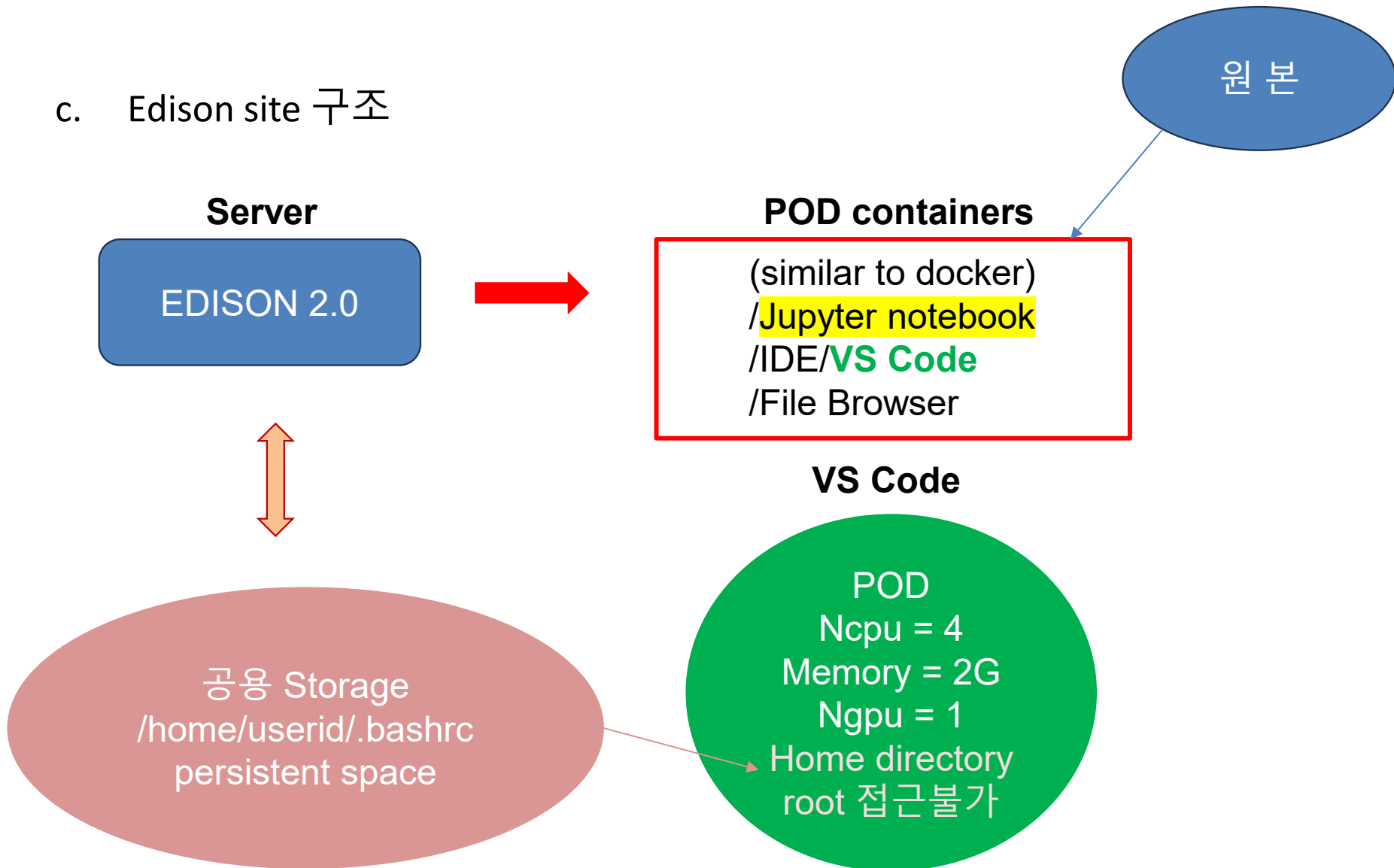
b. 다양한 앱 활용

- Crystal Viewer
 - 물질 구조, 전자 밀도 (surface cut/projection), Brillouin Zone
 - Construction, modification, etc
- Jupyter Notebook
 - Jupyter notebook, terminal 가능 (job submit 안 됨)
- File Browser
 - 고용량 파일 전송
- VS Code → 여름학교 수업 진행
 - Job submit 가능

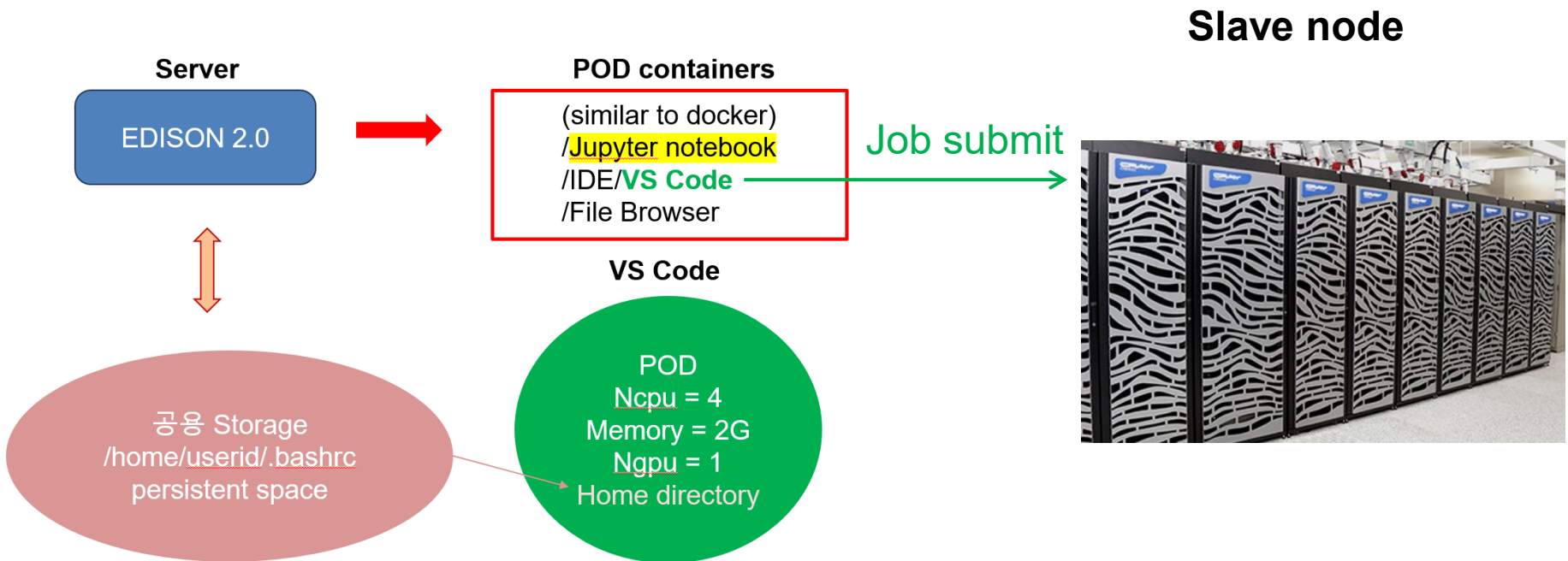
1. Edison homepage

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c. Edison site 구조



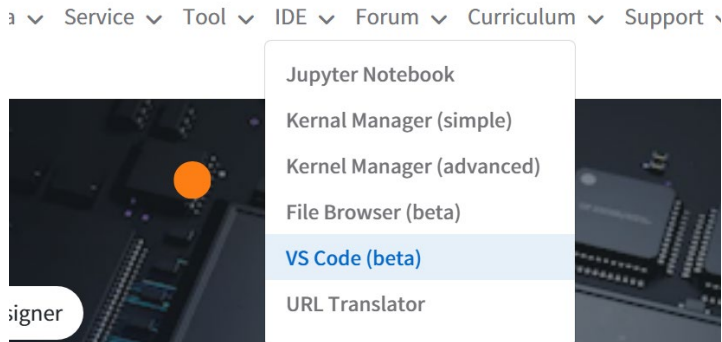
c. Edison site 구조



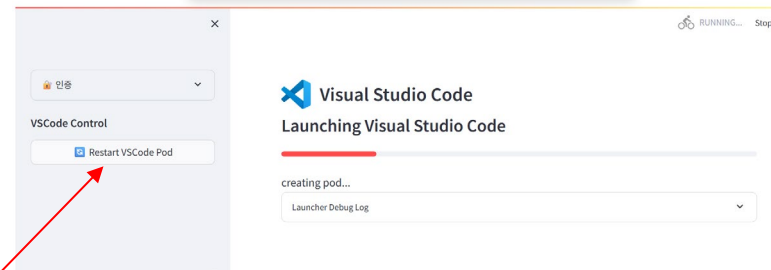
a. 실행

- Waiting until ready: red line to the full
 - 바로 열기 (준비되면 클릭)
- Opening warning: “restricted mode is intended ...”
 - Check “trust parents folder”

Click VS Code (beta)



VSCode Control



2. Log in



VS Code Error

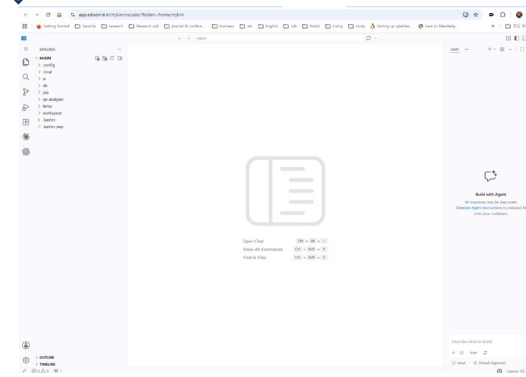
403 Forbidden

nginx

1. POD restart



바로열기 click

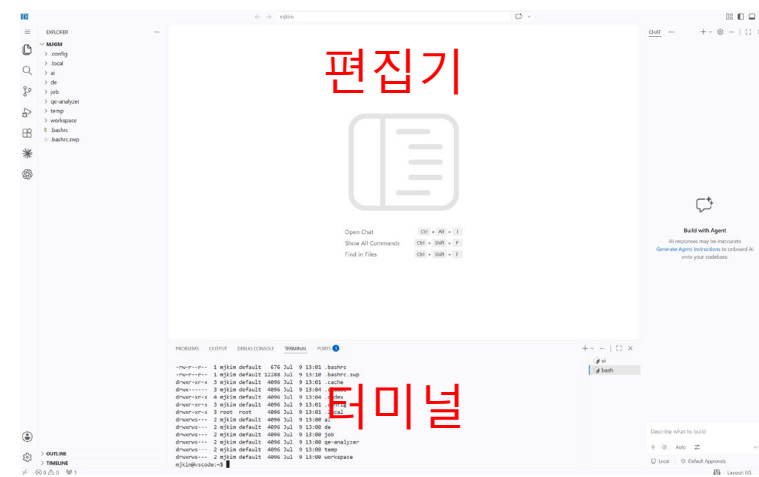
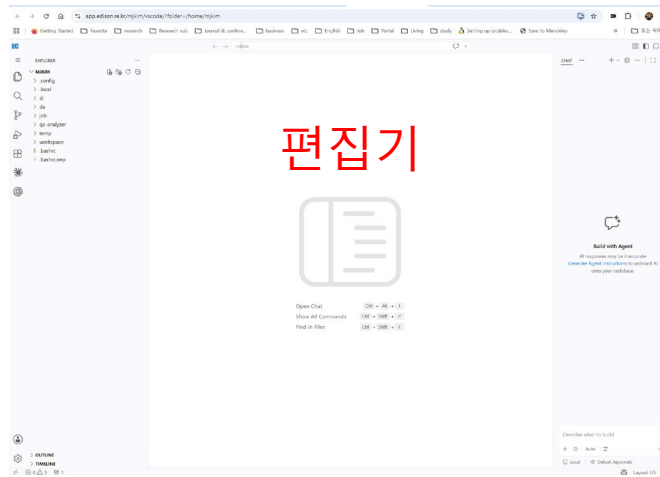


2. VS Code

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b. Panel

- Menu (hamburger icon): 
- Terminal 열기와 화면 분할



PC (ssh client)

MobaXterm
PuTTY

a. Linux 환경

- X-window vs Terminal
 - terminal: cli (command line interface)
- SHELL as operating system
 - echo **\$SHELL** → /bin/bash
- Root vs user
 - / # root by root manager
 - Many other ...
 - /home/userid # User Home
- .bashrc # hidden, in case bash
 - last running script in booting
 - 초기 설정: 터미널 열 때마다
- 파일 제거 시 재생 안 됨

b. Basic command

- **ls** # list, 폴더 보기
- **which ls** # find command
- **echo \$var** # write
- (continued)

c. Important Env variable & setting

- variables
 - SHELL
 - bash 이면 .bashrc가 실행됨
 - PATH
 - 오직 PATH에 잡혀 있는 실행 파일만 실행됨
 - Setting in .bashrc
 - export PATH=\$PATH:new_dir
 - PYTHONPATH
 - 여기에 잡혀 있는 모든 python 모듈들은 import 될 수 있음.
 - cf. 만일 module과 running script가 같은 dir에 있으면 둘 다 잡아줘야 됨.
 - PS1 # prompt setting
- Related command
 - **export** var # var 을 광역변수
 - **source** script # bash script 실행
 - source .bashrc

d. Development tool (command)

- **git** # git 저장소와 연결
 - git clone repo_address
 - git pull # inside git repo
- **conda 설치 (Edison 에서)**
 - conda init # write .bashrc
 - source .bashrc # conda 실행
 - cf
 - conda deactivate
 - conda activate
- **sbatch** # job을 queue에 넣음

b. Basic command

- **ls** # list, 폴더 보기
 - -a (all), -l (long) # options
- **pwd** # present working dir
- **which** # find command
- **echo** # write

.bashrc 세팅

✓ dir, file 종류 등 색 구별

alias ls='ls --color=auto'

✓ prompt 색을 바꾸고, dir 표기

export

PS1="\[\e[32m\]\u@\h:\[\e[34m\]\w\[\e[0m\] \ \$"

Variable setting in bash

no space at both sides of '='

a=3

b='hello'

b. Basic command

- pwd # show present working dir
- ls # list, 폴더 보기
 - -a (all), -l (long) # options
- cd # change directory
- mkdir # make directory
- echo # write
- which # find command
- rm # remove 파일-glob 이용
 - -r (recursive) # dir 까지 다 지움
- history # 지금까지 명령어 기록
- Jobs # 현재 실행중인 job
 - Run job background
 - ctrl + C In case
 - Job running foreground
 - Cannot recover prompt

Linux special dir name

- . # present directory
- .. # parent directory
- ~ # HOME directory

Linux glob symbol

(많은 파일 이름에 matching)

- * # 모든 기호에 해당
- ? # 한 기호에 해당
- [] # 중 하나에 해당

a. git

- \$HOME/
 - `git clone https://github.com/YHKlab/ESC2026.git`
 - `# github.com/yhklab → repo → ESCSchool → copy`
- \$HOME/ESC2026
 - `git pull` `# 수시로 자료가 update 되면 download`

b. Work directory

- `cp -r ESCSchool/exe1-atom exe1`
- `cd exe1`
 - Here is work directory for exe1

c. Python module 설치

- 현재 설치된 pip을 사용할 경우 홈에도 설치가 안 됨 → error
 - which pip
 - `/opt/miniconda3/bin/pip`
 - `pip install wannierberri` → can't write `$HOME/.local/lib` (permission error)
- Install pip in new env
 - `conda create -n myenv` # python version, module, etc
 - `$HOME/.conda/envs/myenv`
 - `conda activate myenv` # still path to `/opt/miniconda3`
 - `(myenv) conda install pip`
 - `(myenv) which pip`
 - `$HOME/.conda/envs/myenv/bin/pip`
 - `$HOME/.conda/envs/myenv/bin/python`
 - `python --version`
- Use myenv to install modules
 - `pip install wannierberri`
 - `python -c "import wannierberri"`
 - `pip install pyfftw`

d. atom